

Introduction to Software Engineering

COSC 336

ReSource (Phase II)

Feasibility Study

Intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle
Management System

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Introduction

Concept

ReSource is an intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System designed to support University's asset management department in identifying, sharing, and reuse extra/underutilized assets such as furniture, equipment, vehicles, and books. By doing so a department that requests an asset could use another department's surplus asset and avoid wasteful spending and storage. Furthermore, the system is designed to track each asset's life-cycle from its registration, maintenance, transfer, and finally when it's donated/recycled or disposed of. The system combines traditional asset management with AI features such as semantic matching, automated classification, and sustainability driven recommendations.

Document Purpose

This document's purpose is to serve as the phase 1 deliverable which highlights the initial project plan and requirement gathering for ReSource. This document will be the foundation of our project that we will turn to as a reference point to verify that we meet all the requirements stated.

Project Overview

High-Level System Description

ReSource is foundationally built on two layers, a traditional asset management layer and an AI-Powered layer.

Layer 1 Classic Asset Management

Module	Function
Asset Registration	Captures structured asset data such as name, category, condition, quantity, location, estimated value, and images

Resource Marketplace	Publishes registered assets as searchable/filterable listings visible to other departments
Request & Reservation	Lets a department submit a request for an asset specifying purpose, quantity, and urgency
Approval & Transfer	Routes accepted requests to an administrator on approval, updates ownership, custody, and location
Life-Cycle Tracking	Logs maintenance, repair, donation, recycling, and disposal events in the asset's history
Access Management	Assigns role-based permissions, governing which actions each user can perform

Layer 2 AI-Powered Features

Module	Function
AI Matching Engine	Ranks open requests against available assets using semantic similarity plus condition, location, and urgency as inputs
AI Classification	Suggests a category and tags at the point of registration, based on the asset's description
Sustainability Optimizer	Outputs a recommended end-of-life action for the asset whether its reuse, repair, donate, recycle, or dispose
LLM Assistant	Accepts natural language questions and maps them to a search/request action
Generative Reporting	Produces a text summary of a selected time period's activity like transfers, savings, and waste diversions from the dataset

Background and Context

Universities routinely purchase new furniture, computers, laboratory equipment, and office materials, while similar usable items remain unused, stored, or thrown out by other departments. This is often not caused by a shortage of resources, but by the absence of an integrated system to identify, match, and track them.

Nowadays, when a department has surplus assets or needs resources, it's typically handled informally and without a shared system. A department that no longer needs an asset stores it in a stockroom, reports it to facilities for disposal, or occasionally informs other departments by email or word of mouth, without any central listing of what is available. When a department needs a certain asset, it will make a request without checking if another

department already has something suitable in surplus. Approvals for transferring assets, when they happen at all, are done through emails or paper forms, with no standardized workflow. Maintenance and repair records, when kept, are usually stored in each department's own spreadsheets or paper logs, making it difficult to see an asset's full history. Decisions about whether to donate, recycle, or dispose of an asset are made case by case, with no consistent criteria and no tracking of the financial and environmental impact.

This manual process of managing assets is unreliable and wastes a significant amount of time and money that could be saved through better coordination. ReSource is designed to replace this outdated, wasteful process with a single, structured platform.

Business Goals and Objectives

- Reduce unnecessary purchases by connecting departments with existing surplus assets.
- Extend the life of campus assets through better tracking and maintenance.
- Increase reuse, donation, and recycling rates across the university.
- Provide insights into the environmental and financial impact of the circular resource management.
- Demonstrate how classic and AI-enhanced features can be combined in a single system.

Expected Benefits

- Financial: Reduced spending by reusing surplus assets and lowering storage costs.
- Operational: clearer accountability through approval workflows and easy to review existing assets.
- Environmental: Reduced waste through donations, recycling, and reusing of assets.
- Organizational: A single source for asset location, condition, and history replacing scattered and informal communication.

Assumptions

- The university would be willing to adopt a centralized platform for internal asset exchange if one were available.
- Departments are willing to accurately register and update the status of their assets.
- Users will have basic computer literacy sufficient to use a web-based system without extensive training.

Project Constraints

- Timeline: The project must be completed within the Fall 2026 semester, following the official Phase 1-8 schedule (September to December 2026).
- Team size: The project is developed by a team of 4 students, with work distributed across planning, requirements, design, and development tasks.
- Scope: As an academic prototype, the system will demonstrate core functionality rather than a production-ready, university-wide deployment.

- Technical resources: AI features (matching, classification) will rely on existing libraries, pretrained models, or APIs appropriate for a semester-length project, rather than custom trained machine learning models.
- Data availability: Real University asset data is not accessible for this course project, so sample or simulated data will be used for development and testing.

Project Scope

In scope	Out of scope
Core classic modules: asset registration, resource marketplace, request/reservation, approval and transfer workflows, life-cycle history, maintenance/repair logging, and role-based user management	Integration with real university procurement or financial systems
Core AI-enhanced modules: AI-assisted classification, AI-based semantic matching with a basic explanation of recommendations, and a sustainability recommendation feature	Full production-level security certification or compliance auditing
Basic operational and AI-generated reporting	Large-scale, continuously trained AI models; AI components will use existing tools/APIs rather than custom model training

note: a basic natural-language query feature may be included as time allows, but is not a core deliverable

Stakeholders

As stated above, the system will support campus resource reuse, asset tracking, and sustainable decisions. This project involves many users with different authorizations and access limits. Therefore understanding their responsibilities and expectations help in defining suitable features, information access, and authorization control.

The table below identifies all the possible stakeholders, starting from the system developing team to all the different end-users. The table clearly states their roles/responsibilities and their needs/expectations to help build up the system.

Stakeholder (Category)	Role/Responsibility	Needs/Expectations & Data Needs	Feature Requirement Mapping	Permissions & Approval Decisions
Administrators (User)	Manage operational workflows, resolve allocation conflicts, approve transfers or end-of-life actions	Approval queues, operational reports, records of rules and changes; needs visibility into all pending requests and transfer records	Final Approval and Transfer Management with User and Access Management	Approve/reject asset transfers, end-of-life actions, and manage user roles
Procurement Team (User)	Check stock before purchasing and accept or reject requests	Inventory and demand visibility, compatibility information, purchase records; needs inventory levels and purchase-request history	AI Matching Engine giving AI-Generated Reports	View inventory and requests and approve whether a new purchase is justified or if we can reuse surplus assets
Department Representative (User)	Identify department needs and surplus assets with coordinated requests and transfers	Asset inventory and request status since they need visibility into their own department's inventory and requests	Request & Reserve New Asset	Create/edit own department's listings and submit requests while approval sits with Administrators
Asset Coordinators (User)	Register assets and maintain condition, ownership, and location records plus verify availability and review suggested matches	Accurate asset histories, current availability, clear specifications, relevant matches, status updates	Asset Registration & Publication, AI Matching, Sustainability Optimization Resource Marketplace, and Life-Cycle Tracking	Register/update assets and confirm receipt/condition on completed transfers
University Management (Customer)	Represent the university, confirm project priorities and budget authority, sign off on major decisions	Accurate asset records, reduced spending, clear high-level reporting with needs summary-level cost and impact data, not operational detail	Generative AI Reporting (financial savings, waste diversion, avoided purchases)	Final authority on budget and adoption without day-to-day operational permissions
Developers (Software Developer)	Gather/understand requirements, design, build, and test the system	Clearly stated requirements and scope, defined interfaces, access to development tools/environment	Not feature-specific because they are responsible for implementing all classic and AI modules	Full read/write access to codebase and dev environment without business-decision approval authority

Lab Instructors (Development Manager)	Provide academic guidance, review phase deliverables, assess progress against course requirements	Regular, clear deliverables and documentation visibility into evaluation criteria and rubric alignment across phases	Not feature-specific since they governs project process and academic evaluation, not system functionality	Approve/grade phase deliverables without permissions within the deployed system itself
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These stakeholder needs will guide the system’s development and evaluation.

Market Analysis

Technical Analysis

1) Hardware and Software Requirements

ReSource is proposed as a web-based system that combines campus asset management with AI-assisted resource matching, classification, and sustainability recommendations.

Development will use free and open-source tools, pretrained models, or APIs, consistent with the scope defined in Phase 1.

- **Backend (Python/Django or Java):** The backend would process asset registrations, requests, reservations, approvals, transfers, and maintenance records. Django provides user authentication of accounts, groups, permissions, and password hashing [1].
- **Frontend (HTML, CSS, and JavaScript):** The interface would provide asset listings, search filters, registration forms, request tracking, and dashboards appropriate to each user's role.
- **Database (MySQL) :** Database would store users, departments, assets, requests, transfers, and life-cycle records. Linking these records would support ownership tracking and reporting. MySQL is supported by Django as well.
- **AI Libraries:** Use pre-defined AI libraries such as (Sentence Transformers) for semantic search, comparing asset and description. LLM could also be used to support generated reports and natural-language interaction
- **Hardware:** Existing teams' and labs computers would be used for development and testing.

For example, when a department needs equipment, a user would enter the request through the frontend (HTML, CSS, and JavaScript). The backend (Python/Django or Java) would process the request and retrieve available asset records from the MySQL database. Sentence Transformers would compare the request's meaning with asset descriptions to help identify suitable matches. The backend would apply checks for availability and required specifications, and the frontend would display the results.

When the user selects an asset, the backend would verify permissions and quantities, record the reservation in MySQL, and route the request for approval. After the transfer is confirmed, it would update the asset's custody, location, and life-cycle history in the database. An LLM could subsequently generate a summary using the recorded transfer information.

2) Skills Required

The project requires the following steps:

- **Software Engineering skills:** Designing/Planning, Implementing, and testing skills
- **Web Development:** UI, and backend workflows. Making the system easy to use.
- **Database Design:** Connecting assets, user, request, and maintenance records while maintaining accurate quantities and ownership.
- **AI integration** (layer 2 of the system): Using pretrained models or APIs, preparing input data, evaluating recommendations, and handling unsuccessful responses.

- **Collaboration and deployment:** Managing shared code through GitHub, integrating modules, and configuring the demonstration environment.

For example: One team member develops the interface, another builds the backend and approval workflows, a third designs the database for asset and transfer records, and the fourth integrates AI matching and reporting. All four members contribute to testing and combining the modules.

3) Feasibility

- With these tools and skills, the project is possible within the proposed timeline
- Open-source software means no big costs.
- Team members already have the basic skills needed, and extra parts (like AI) can be learned during the project.

As real asset data is unavailable, development and evaluation will use sample or simulated records. Testing will follow the Phase 1 success criteria, including the planned workload of 100 asset records and five simultaneous users. AI accuracy and response-time targets remain to be demonstrated.

Overall, ReSource appears technically feasible as a prototype, provided the team confirms the required skills, validates the selected AI tools, and maintains the agreed scope.

Financial Analysis

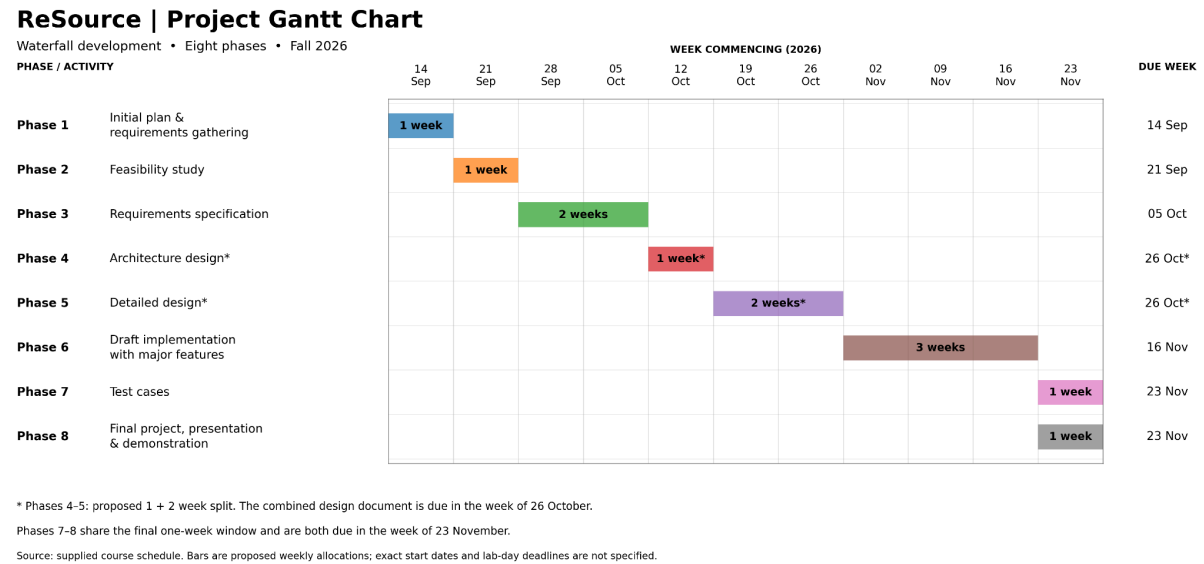
Legal and regulatory analysis

Risk assessment

Project schedule

Phase	Main Activities	Milestone	Dependencies
Phase 1 Planning and Requirements Gathering	Define the scope, stakeholders, initial requirements, constraints, and preliminary project plan.	Initial plan and requirement gathering document completed. <i>Due: week of 14 Sep 2026</i>	None. Project initiation.
Phase 2 Feasibility Study	Assess technical, economic, operational, and schedule feasibility, including AI tools and data availability.	Feasibility document completed. <i>Due: week of 21 Sep 2026</i>	Phase 1: Scope and initial requirements.
Phase 3 Requirements Specification	Finalize functional, non-functional, AI, and sustainability requirements, business rules, and acceptance criteria.	Requirements document completed. <i>Due: week of 5 Oct 2026</i>	Phases 1 and 2: Stakeholder needs and feasibility findings.
Phase 4 Architecture Design	Define system layers, modules, interfaces, data flow, and the AI integration approach.	Architecture section completed for the combined design document. <i>Due: week of 26 Oct 2026</i>	Phase 3: Completed requirements specification.
Phase 5 Detailed Design	Specify database schemas, workflows, UML models, user interfaces, and component-level logic.	Combined architecture and detailed design document completed. <i>Due: week of 26 Oct 2026</i>	Phases 3 and 4: Requirements and system architecture.

Phase 6 Implementation	Develop core asset-management functions and planned AI features; integrate major system modules.	Draft implementation with major features completed. <i>Due: week of 16 Nov 2026</i>	Phases 4 and 5: Completed architecture and detailed design.
Phase 7 Testing and Test Cases	Prepare and execute functional, integration, usability, security, and AI test cases; document results and corrections.	Test-case details completed. <i>Due: week of 23 Nov 2026</i>	Phases 3 and 5: Test basis. Phase 6: Prototype for test execution.
Phase 8 Final Documentation and Delivery	Finalize documentation and corrections; prepare the project presentation and system demonstration.	Final project submitted; presentation and demonstration delivered. <i>Due: week of 23 Nov 2026</i>	Phases 6 and 7: Working prototype, test results, and necessary corrections.



Recommendations

Constraints

Constraint	Why it matters	Simple response
Legal	The system must follow applicable laws when handling university data and AI services.	Use only approved data and services; check legal requirements before deployment.
Ethical	AI recommendations should not unfairly favor departments or make important decisions by itself.	Keep humans responsible for approvals and allow users to review or override AI suggestions.
Privacy	Asset records may include user, department, location, financial, or operational information.	Collect only needed data and limit who can view sensitive information.
Security	Unauthorized access or changes could affect asset ownership, transfers, and records.	Use authentication, role-based access, secure storage, and audit logs.
Procurement	External AI tools, cloud services, or software may require university approval or payment.	Prefer approved tools and confirm cost, licensing, and vendor requirements early.
Ownership	The system tracks who owns or controls each asset and records transfers between departments.	Require authorized approval before changing ownership, custody, or location.
Safety	Some assets may be damaged, unsafe, or unsuitable for reuse or transfer.	Require inspection or maintenance checks before risky assets are reused or transferred.
University Policy	The project must follow university rules for IT, data, assets, procurement, and disposal.	Align workflows and permissions with university procedures and responsible staff.

References

[1] “User authentication in Django | Django documentation,” *Django Project*.
<https://docs.djangoproject.com/en/6.1/topics/auth/>